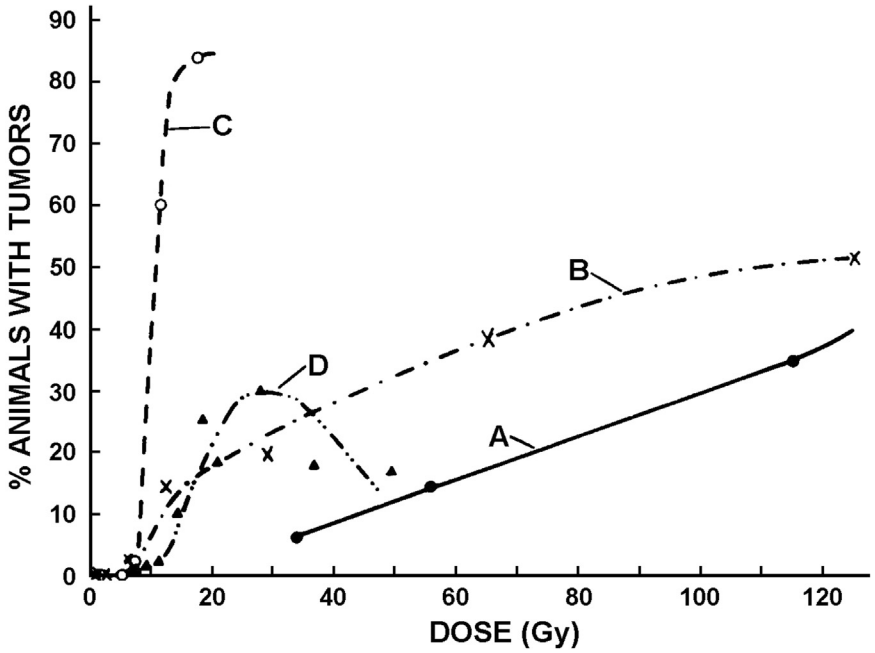


Figure 15.8 Tumor frequency as a function of absorbed dose; see text.

1968). Figure 15.8 seems to indicate a threshold dose at $\leq$5 Gy below which no effect is observed. However, some animal experiments show what looks like a small beneficial effect at small radiation doses, see Figure 15.12.

15.6.4. Mental retardation

In Hiroshima and Nagasaki, both severe mental retardation and lower intelligence levels (obtained from intelligence tests) occurred following prenatal exposure during the 8–15 weeks after pregnancy. In the following 16–25 weeks after ovulation, lesser vulnerability was observed. The exposure to 1 Gy during the early period increased the frequency of mental retardation 40% (normal frequency about 0.8%), and lowered IQ by 25–30 points. No cases of severe mental retardation has been observed at exposures $<$0.5 Gy. No radiation effects on the brain have been observed at low dose rates.

15.7. RADIATION STERILIZATION

Figure 15.2 and Table 15.5 illustrate that at appropriate dosages radiation is an effective means of destroying higher organisms with less harm to simpler ones. This has a practical consequence because radiation can be used for the conservation of food in a quite different manner than the conventional methods of heat, canning, and freezing.

Conservation by radiation attempts to achieve the complete destruction of all bacteria with minimum change in the taste of the food (due to the formation of small amounts of decomposition products). Radiation pasteurization (i.e. partial sterilization with lower doses) and irradiation at low temperatures cause correspondingly smaller taste changes.

The safety of food irradiation has been accepted by IAEA, FAO (Food and Agricultural Organization) and WHO. It has been used for several decades without any negative health effects. More than 35 countries have approved some 50 different irradiated food products for unrestricted human consumption. Examples of the foods are (country, maximum dose): potatoes (Canada, 100 Gy), onions (Israel, 100 Gy), shrimp (Netherlands, 1000 Gy), fried meat in plastic bags (Soviet Union, 8000 Gy), and wheat and wheat products (USA, 500 Gy).

Medical supplies, which must be sterile, can be manufactured and packed by conventional techniques, after which the packages may be exposed to high energy penetrating radiation (e.g. ^{60}Co, ^{137}Cs or electrons from small accelerators). The radiation kills all bacteria with little damage to low molecular weight compounds. In this common technique, the irradiation source is quite large (0.1−1 MCi; around 10^{11}Bq). The packages slowly pass through the source, such that the doses are on the order of 10−30 kGy. The same facilities can be used also for other purposes such as food sterilization.

Radiation doses of ~2 Gy produce sterility in the human male for about 12 months while higher doses lead to permanent sterility; libido and potency are unchanged. Permanent sterility occurs in 60% of young women given 8 Gy of fractionated irradiation to the ovary, but occasional pregnancies can occur after doses of up to 5−8 Gy; the children are apparently normal at birth (BEIR V).

Radiation has been used to sterilize (80 Sv) the males of certain insect species which when released (the ratio between the sterilized and untreated males was approximately 4:1), mate with females and prevent further reproduction of the species. This technique has been used in the US, Mexico, Egypt, Libya, etc to eradicate screw worm flies which can cause huge damage to the cattle industry. The technique is now also used against other insect species which are threats to agriculture.

15.8. GENETIC EFFECTS

In 1927, Muller showed how irradiation of fruit flies (Drosophila M.) could produce new species that were defective with regard to the parents (e.g. lacked wings). This defect was carried in the genes to later generations. This dramatic demonstration of the mutagenic effect of radiation has been extended to other primitive species. The vast majority of such changes are recessive, i.e. in order to become dominant both parents must carry the same DNA-damage. From all experience as well as the theoretical considerations described above, the likelihood for two identical DNA impairments to